

3.5 Preliminary Product Description

3.5.1 Overview

The **Smart Event Management and Ticket Booking System** is a web-based application designed to simplify the process of organizing events and booking tickets online. The system provides a centralized platform where administrators can create, update, and manage events, while users can register, browse available events, book tickets, make payments, and submit feedback after attending events.

The application eliminates the limitations of manual event management by providing an efficient, secure, and user-friendly solution. It improves communication between event organizers and participants while maintaining accurate records of bookings, payments, and user feedback.

3.5.2 Product Features

The proposed system provides the following features:

- User Registration and Login
 - Secure Authentication
 - Event Creation and Management
 - Event Browsing and Search
 - Online Ticket Booking
 - Payment Processing
 - Booking History Management
 - User Profile Management
 - Feedback and Rating System
 - Administrative Dashboard
 - Report Generation
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3.5.3 User Characteristics

The system is designed for two categories of users:

Administrator

The administrator has complete control over the application. The administrator can:

- Manage user accounts.

- Create, update, and delete events.
- Monitor ticket bookings.
- View payment records.
- Review user feedback.
- Generate reports.

User

A registered user can:

- Register and log in.
 - Browse available events.
 - View event details.
 - Book event tickets.
 - Make online payments.
 - View booking history.
 - Submit event feedback.
 - Manage personal profile.
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3.5.4 Product Perspective

The Smart Event Management and Ticket Booking System is developed as a standalone web application that can be accessed using any modern web browser. The application follows a client-server architecture where the frontend provides the user interface, the backend processes business logic, and the MongoDB database stores application data securely. The system supports multiple users simultaneously and provides real-time access to event information.

3.5.5 Benefits of the Proposed System

The proposed system offers several advantages:

- Reduces manual work in event management.
- Provides quick and secure ticket booking.
- Improves data accuracy and consistency.
- Enables online payment processing.
- Simplifies event administration.
- Provides better user experience.
- Maintains centralized event records.
- Generates reports for administrative purposes.
- Supports efficient communication between administrators and users.

3.5.6 Limitations

The current version of the system has the following limitations:

- Internet connectivity is required.
- Online payment depends on third-party payment services.
- Only registered users can book tickets.
- Advanced analytics and notification services are not included in the current version.

3.5.7 Future Enhancements

The system can be enhanced in the future by implementing:

- Email and SMS notifications.
- QR Code-based ticket verification.
- Mobile application support.
- Event recommendation system.
- Multiple payment gateway integration.
- Digital ticket generation.
- Event analytics dashboard.
- Multi-language support.